Gull Tracker API

1.0.0 OAS 3.1

/openapi.json

Gull Migration patterns and corosponding weather data

health

GET `/api/v1/health` Health Check

Parameters Try it out

No parameters

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls `Accept` header. **Example Value** Schema `"string"`	*No links*

gulls

GET `/api/v1/gulls/{gull_id}/route-with-weather` Get Gull Route With Weather

Parameters

Try it out

Name	Description
gull_id * required integer (path)	gull_id

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. **Example Value** Schema	*No links*
422	Validation Error	*No links*


```
[
  {
    "id": 0,
    "gull_id": 0,
    "recorded_at": "2026-03-20T21:25:31.653Z",
    "latitude": 0,
    "longitude": 0,
    "event_id": "string",
    "sensor_type": "string",
    "visible": "string",
    "temperature_c": 0,
    "precipitation_mm": 0,
    "wind_u": 0,
    "wind_v": 0,
    "surface_pressure": 0,
    "weather_id": 0,
    "weather_observed_at": "2026-03-20T21:25:31.653Z"
  }
]
```

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


GET `/api/v1/gulls/` List Gulls

Parameters Try it out

Name	Description
species `string | (string | null)` *(query)*	species

Responses

Code	Description	Links
200	Successful Response Media type **application/json** Controls `Accept` header.	*No links*

Example Value Schema


```
[
  {
    "tag_id": "string",
    "species": "string",
    "common_name": "string",
    "study_name": "string",
    "id": 0
  }
]
```


422 Validation Error *No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


POST `/api/v1/gulls/` Create Gull

Parameters **Try it out**

No parameters

Request body required **application/json**

Example Value Schema


```
{
  "tag_id": "string",
  "species": "string",
  "common_name": "string",
```

```
  "study_name": "string"
}
```

Responses

Code	Description	Links
201	Successful Response Media type **application/json** Controls `Accept` header. **Example Value** Schema	*No links*
422	Validation Error Media type **application/json** **Example Value** Schema	*No links*

```
{
  "tag_id": "string",
  "species": "string",
  "common_name": "string",
  "study_name": "string",
  "id": 0
}
```

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

GET /api/v1/gulls/{gull_id} Get Gull

Parameters

Try it out

Name	Description
gull_id * required integer (path)	gull_id

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value Schema	No links
422	Validation Error Media type application/json Example Value Schema	No links

```
{
  "tag_id": "string",
  "species": "string",
  "common_name": "string",
  "study_name": "string",
  "id": 0
}
```

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
```

```
            ],
            "msg": "string",
            "type": "string",
            "input": "string",
            "ctx": {}
          }
        ]
      }
```


PUT `/api/v1/gulls/{gull_id}` Update Gull Full

Parameters Try it out

Name	Description
gull_id * required `integer` *(path)*	gull_id

Request body required application/json

Example Value Schema


```
{
  "tag_id": "string",
  "species": "string",
  "common_name": "string",
  "study_name": "string"
}
```


Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header.	*No links*

Example Value Schema


```
{
  "tag_id": "string",
  "species": "string",
  "common_name": "string",
  "study_name": "string",
  "id": 0
}
```


422 Validation Error *No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


PATCH `/api/v1/gulls/{gull_id}` Update Gull Partial

Parameters **Try it out**

Name	Description
gull_id * required `integer` *(path)*	gull_id

Request body required **application/json**

Example Value Schema

```
{
  "tag_id": "string",
  "species": "string",
  "common_name": "string",
  "study_name": "string"
}
```


Responses

Code	Description	Links
200	Successful Response Media type **application/json** Controls `Accept` header. **Example Value** Schema	*No links*
422	Validation Error Media type **application/json** **Example Value** Schema	*No links*


```
{
  "tag_id": "string",
  "species": "string",
  "common_name": "string",
  "study_name": "string",
  "id": 0
}
```

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
```

```
}
```

DELETE `/api/v1/gulls/{gull_id}` Delete Gull

Parameters

Try it out

Name	Description
gull_id * required `integer` *(path)*	gull_id

Responses

Code	Description	Links
204	Successful Response	*No links*
422	Validation Error	*No links*

Media type

application/json

Example Value Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

GET `/api/v1/gulls/{gull_id}/movement-summary`

Get a movement summary for a gull

Parameters

Try it out

Name	Description
gull_id * required `integer` *(path)*	gull_id

Responses

Code	Description	Links
200	Successful Response Media type **application/json** Controls `Accept` header. **Example Value** Schema	*No links*
422	Validation Error Media type **application/json** **Example Value** Schema	*No links*

```
{
  "gull_id": 0,
  "tag_id": "string",
  "species": "string",
  "total_trackpoints": 0,
  "first_recorded_at": "2026-03-20T21:25:31.660Z",
  "last_recorded_at": "2026-03-20T21:25:31.660Z",
  "duration_hours": 0,
  "total_distance_km": 0,
  "average_step_distance_km": 0,
  "average_temperature_c": 0,
  "average_precipitation_mm": 0
}
```

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


trackpoints

GET `/api/v1/trackpoints` List Gull Trackpoints

Parameters

Try it out

Name	Description
gull_id integer | (integer | null) (query)	gull_id

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value Schema	*No links*

```
[
  {
    "gull_id": 0,
    "recorded_at": "2026-03-20T21:25:31.661Z",
    "latitude": -90,
    "longitude": -180,
    "event_id": "string",
    "sensor_type": "string",
    "visible": "string",
    "id": 0
  }
]
```


422 Validation Error *No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


POST `/api/v1/trackpoints` Create Gull Trackpoint

Parameters **Try it out**

No parameters

Request body required **application/json**

Example Value Schema


```
{
  "gull_id": 0,
  "recorded_at": "2026-03-20T21:25:31.662Z",
```

```
  "latitude": -90,
  "longitude": -180,
  "event_id": "string",
  "sensor_type": "string",
  "visible": "string"
}
```


Responses

Code	Description	Links
201	Successful Response Media type **application/json** Controls `Accept` header. **Example Value** Schema	*No links*
422	Validation Error Media type **application/json** **Example Value** Schema	*No links*


```
{
  "gull_id": 0,
  "recorded_at": "2026-03-20T21:25:31.663Z",
  "latitude": -90,
  "longitude": -180,
  "event_id": "string",
  "sensor_type": "string",
  "visible": "string",
  "id": 0
}
```

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
```

```
            "ctx": {}
          }
        ]
      }
```


GET `/api/v1/trackpoints/{trackpoint_id}` Get Gull Trackpoint

Parameters

Try it out

Name	Description
trackpoint_id * required `integer` *(path)*	trackpoint_id

Responses

Code	Description	Links
200	Successful Response Media type **application/json** Controls `Accept` header. **Example Value** Schema	*No links*
422	Validation Error	*No links*


```
{
  "gull_id": 0,
  "recorded_at": "2026-03-20T21:25:31.664Z",
  "latitude": -90,
  "longitude": -180,
  "event_id": "string",
  "sensor_type": "string",
  "visible": "string",
  "id": 0
}
```

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


PUT `/api/v1/trackpoints/{trackpoint_id}` Update Gull Trackpoint Full

Parameters Try it out

Name	Description
trackpoint_id * required integer (path)	trackpoint_id

Request body required **application/json**

Example Value Schema


```
{
  "gull_id": 0,
  "recorded_at": "2026-03-20T21:25:31.665Z",
  "latitude": -90,
  "longitude": -180,
  "event_id": "string",
  "sensor_type": "string",
  "visible": "string"
}
```

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type

application/json

Controls `Accept` header.

Example Value Schema


```
{
  "gull_id": 0,
  "recorded_at": "2026-03-20T21:25:31.665Z",
  "latitude": -90,
  "longitude": -180,
  "event_id": "string",
  "sensor_type": "string",
  "visible": "string",
  "id": 0
}
```


422	Validation Error	*No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


PATCH `/api/v1/trackpoints/{trackpoint_id}` Update Gull Trackpoint Partial

Parameters

Try it out

Name	Description
trackpoint_id * required integer (path)	trackpoint_id

Request body required

application/json

Example Value Schema


```
{
  "gull_id": 0,
  "recorded_at": "2026-03-20T21:25:31.666Z",
  "latitude": -90,
  "longitude": -180,
  "event_id": "string",
  "sensor_type": "string",
  "visible": "string"
}
```


Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. **Example Value** Schema	*No links*


```
{
  "gull_id": 0,
  "recorded_at": "2026-03-20T21:25:31.667Z",
  "latitude": -90,
  "longitude": -180,
  "event_id": "string",
  "sensor_type": "string",
```

```
  "visible": "string",
  "id": 0
}
```


422	Validation Error	No links

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


DELETE `/api/v1/trackpoints/{trackpoint_id}` Delete Gull Trackpoint

Parameters Try it out

Name	Description
trackpoint_id * required `integer` *(path)*	trackpoint_id

Responses

Code	Description	Links
204	Successful Response	*No links*

Code	Description	Links
422	Validation Error	No links

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


GET `/api/v1/trackpoints/{trackpoint_id}/weather`

Get the best matching weather observation for a trackpoint

Parameters Try it out

Name	Description
trackpoint_id * required `integer` *(path)*	trackpoint_id

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls `Accept` header.

Example Value Schema


```
{
  "trackpoint_id": 0,
  "gull_id": 0,
  "weather_id": 0,
  "trackpoint_recorded_at": "2026-03-20T21:25:31.668Z",
  "weather_observed_at": "2026-03-20T21:25:31.668Z",
  "time_difference_minutes": 0,
  "distance_km": 0,
  "match_method": "string",
  "latitude": 0,
  "longitude": 0,
  "temperature_c": 0,
  "precipitation_mm": 0,
  "wind_u": 0,
  "wind_v": 0,
  "surface_pressure": 0
}
```


422 Validation Error *No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


weather

GET `/api/v1/weather` List Weather

Parameters Try it out

Name	Description
year integer | (integer | null) (query)	year

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value Schema	No links

```
[
  {
    "observed_at": "2026-03-20T21:25:31.670Z",
    "latitude": -90,
    "longitude": -180,
    "year": 0,
    "temperature_c": 0,
    "precipitation_mm": 0,
    "wind_u": 0,
    "wind_v": 0,
    "surface_pressure": 0,
    "source": "string",
    "dataset_name": "string",
    "id": 0
  }
]
```

422	Validation Error Media type application/json Example Value Schema	No links

```
{
  "detail": [
    {
```

```
            "loc": [
              "string",
              0
            ],
            "msg": "string",
            "type": "string",
            "input": "string",
            "ctx": {}
          }
        ]
      }
```


POST `/api/v1/weather` Create Weather

Parameters Try it out

No parameters

Request body required application/json

Example Value Schema


```
{
  "observed_at": "2026-03-20T21:25:31.671Z",
  "latitude": -90,
  "longitude": -180,
  "year": 0,
  "temperature_c": 0,
  "precipitation_mm": 0,
  "wind_u": 0,
  "wind_v": 0,
  "surface_pressure": 0,
  "source": "string",
  "dataset_name": "string"
}
```


Responses

Code	Description	Links
201	Successful Response	*No links*

Media type

application/json

Controls `Accept` header.

Example Value Schema


```
{
  "observed_at": "2026-03-20T21:25:31.671Z",
  "latitude": -90,
  "longitude": -180,
  "year": 0,
  "temperature_c": 0,
  "precipitation_mm": 0,
  "wind_u": 0,
  "wind_v": 0,
  "surface_pressure": 0,
  "source": "string",
  "dataset_name": "string",
  "id": 0
}
```


422 Validation Error *No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


GET `/api/v1/weather/{weather_id}` Get Weather

Parameters Try it out

Name	Description

weather_id * required

integer

(path)

weather_id

Responses

Code	Description	Links
200	Successful Response Media type **application/json** Controls Accept header. **Example Value** Schema	*No links*
422	Validation Error Media type **application/json** **Example Value** Schema	*No links*

```
{
  "observed_at": "2026-03-20T21:25:31.672Z",
  "latitude": -90,
  "longitude": -180,
  "year": 0,
  "temperature_c": 0,
  "precipitation_mm": 0,
  "wind_u": 0,
  "wind_v": 0,
  "surface_pressure": 0,
  "source": "string",
  "dataset_name": "string",
  "id": 0
}
```

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
```

```
                "type": "string",
                "input": "string",
                "ctx": {}
            }
        ]
    }
```


PUT /api/v1/weather/{weather_id} Update Weather

Parameters

Try it out

Name	Description
weather_id * required integer (path)	weather_id

Request body required

application/json

Example Value Schema


```
{
  "observed_at": "2026-03-20T21:25:31.673Z",
  "latitude": -90,
  "longitude": -180,
  "year": 0,
  "temperature_c": 0,
  "precipitation_mm": 0,
  "wind_u": 0,
  "wind_v": 0,
  "surface_pressure": 0,
  "source": "string",
  "dataset_name": "string"
}
```


Responses

Code	Description	Links
200	Successful Response	*No links*

Media type

application/json

Controls `Accept` header.

Example Value Schema


```
{
  "observed_at": "2026-03-20T21:25:31.674Z",
  "latitude": -90,
  "longitude": -180,
  "year": 0,
  "temperature_c": 0,
  "precipitation_mm": 0,
  "wind_u": 0,
  "wind_v": 0,
  "surface_pressure": 0,
  "source": "string",
  "dataset_name": "string",
  "id": 0
}
```


422 Validation Error *No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


PATCH `/api/v1/weather/{weather_id}` Patch Weather

Parameters Try it out

Name	Description

weather_id * required

integer

(path)

weather_id

Request body required

application/json

Example Value Schema


```
{
  "observed_at": "2026-03-20T21:25:31.675Z",
  "latitude": -90,
  "longitude": -180,
  "year": 0,
  "temperature_c": 0,
  "precipitation_mm": 0,
  "wind_u": 0,
  "wind_v": 0,
  "surface_pressure": 0,
  "source": "string",
  "dataset_name": "string"
}
```


Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. **Example Value** Schema	*No links*


```
{
  "observed_at": "2026-03-20T21:25:31.676Z",
  "latitude": -90,
  "longitude": -180,
  "year": 0,
  "temperature_c": 0,
  "precipitation_mm": 0,
  "wind_u": 0,
  "wind_v": 0,
  "surface_pressure": 0,
```

```
  "source": "string",
  "dataset_name": "string",
  "id": 0
}
```


422	Validation Error	No links

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


DELETE `/api/v1/weather/{weather_id}` Delete Weather

Parameters Try it out

Name	Description
weather_id * required `integer` *(path)*	weather_id

Responses

Code	Description	Links
204	Successful Response	No links

422 Validation Error *No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


GET `/api/v1/weather/coverage/summary` Weather Coverage

Parameters Try it out

No parameters

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type

application/json

Controls `Accept` header.

Example Value Schema


```
{
  "available_years": [
    0
```

```
    ],
    "total_records": 0,
    "earliest_timestamp": "2026-03-20T21:25:31.677Z",
    "latest_timestamp": "2026-03-20T21:25:31.677Z"
  }
```


Schemas

GullCreate object

tag_id* string [1, 100] characters

species* string [1, 150] characters

common_name (string | null)

study_name (string | null)

GullMovementSummaryRead object

gull_id* integer

tag_id* string

species* string

total_trackpoints* integer

first_recorded_at* (string | null)

last_recorded_at* (string | null)

duration_hours* number

total_distance_km* number

average_step_distance_km* number

average_temperature_c* (number | null)

average_precipitation_mm* (number | null)

GullRead object

tag_id* string [1, 100] characters

species* string [1, 150] characters

common_name (string | null)

study_name (string | null)

id* integer

GullTrackPointCreate ^ object

gull_id* integer

recorded_at* string date-time

latitude* number [-90, 90]

longitude* number [-180, 180]

event_id > (string | null)

sensor_type > (string | null)

visible > (string | null)

GullTrackPointRead ^ object

gull_id* integer

recorded_at* string date-time

latitude* number [-90, 90]

longitude* number [-180, 180]

event_id > (string | null)

sensor_type > (string | null)

visible > (string | null)

id* integer

GullTrackPointUpdate ^ object

gull_id > (integer | null)

recorded_at > (string | null)

latitude > (number | null)

longitude > (number | null)

event_id > (string | null)

sensor_type > (string | null)

visible > (string | null)

GullUpdate ^ object

tag_id > (string | null)

species > (string | null)

common_name > (string | null)

study_name > (string | null)

HTTPValidationError ^ object

detail > array<object>

TrackpointWeatherMatchRead ^ object

trackpoint_id* integer

gull_id* integer

weather_id* integer

trackpoint_recorded_at* string date-time

weather_observed_at* string date-time

time_difference_minutes* number

distance_km* number

match_method* string

latitude* number

longitude* number

temperature_c > (number | null)

precipitation_mm > (number | null)

wind_u > (number | null)

wind_v > (number | null)

surface_pressure > (number | null)

TrackpointWithWeatherRead ^ object

id* integer

gull_id* integer

recorded_at* string date-time

latitude* number

longitude* number

event_id > (string | null)

sensor_type > (string | null)

visible > (string | null)

temperature_c > (number | null)

precipitation_mm > (number | null)

wind_u > (number | null)

wind_v > (number | null)

surface_pressure > (number | null)

weather_id > (integer | null)

weather_observed_at > (string | null)

ValidationError ^ object

loc* > array<(string | integer)>

msg* string

type* string

input any

ctx object

WeatherCoverageRead ^ object

available_years* > array<integer>

total_records* integer

earliest_timestamp* > (string | null)

latest_timestamp* > (string | null)

WeatherObservationCreate ^ object

observed_at* string date-time

latitude* number [-90, 90]

longitude* number [-180, 180]

year* integer

temperature_c > (number | null)

precipitation_mm > (number | null)

wind_u > (number | null)

wind_v > (number | null)

surface_pressure > (number | null)

source > (string | null)

dataset_name > (string | null)

WeatherObservationPatch ^ object

observed_at > (string | null)

latitude > (number | null)

longitude > (number | null)

year > (integer | null)

temperature_c > (number | null)

precipitation_mm > (number | null)

wind_u > (number | null)

wind_v > (number | null)

surface_pressure > (number | null)

source > (string | null)

dataset_name > (string | null)

Additional properties forbidden

WeatherObservationRead ^ object

observed_at* string date-time

latitude* number [-90, 90]

longitude* number [-180, 180]

year* integer

temperature_c > (number | null)

precipitation_mm > (number | null)

wind_u > (number | null)

wind_v > (number | null)

surface_pressure > (number | null)

source > (string | null)

dataset_name > (string | null)

id* integer

WeatherObservationUpdate ^ object

observed_at* string date-time

latitude* number [-90, 90]

longitude* number [-180, 180]

year* integer

temperature_c > (number | null)

precipitation_mm > (number | null)

wind_u > (number | null)

wind_v > (number | null)

surface_pressure > (number | null)

source > (string | null)

dataset_name > (string | null)